17)

	S	M	L	
Form	67( )	26( )	16( )	109
No Form	128( )	63( )	46( )	237
	195	89	62	346

$E_{S,F} = \frac{109\times195}{346} = 61.43 \quad E_{M,F} = \frac{109\times89}{346} = 28.04 \quad E_{L,F} = \frac{109\times62}{346} = 19.53$

$E_{S,N} = \frac{237\times195}{346} = 133.57 \quad E_{M,N} = \frac{237\times89}{346} = 60.96 \quad E_{L,N} = \frac{237\times62}{346} = 42.47$

$$\chi^2 = \sum \frac{(O_i - E_i)}{E_i} = \frac{(67-61.43)^2}{61.43} + \frac{(26-28.04)^2}{28.04} + \frac{(16-19.53)^2}{19.53} + \frac{(128-133.57)^2}{133.57} + \frac{(63-60.96)^2}{60.96} + \frac{(46-42.47)^2}{42.47}$$

$= 1.784$

$\chi^2_{(r-1)(c-1),\,0.05} = \chi^2_{2,0.05} = 5.991 \Rightarrow$ non reject.

23.

	1 day	1-7 day	7 day	
50000 ↓	38 (32.27)	16 (21.08)	14 (10.65)	68
50000 - 74999	54 (49.07)	26 (28.52)	12 (14.41)	92
75000 - 99999	35 (35.2)	22 (20.46)	9 (10.34)	66
10000 & above	33 (39.47)	29 (22.94)	12 (11.59)	74
	160	93	47	300

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(38-32.27)^2}{32.27} + \frac{(16-21.08)^2}{21.08} + \frac{(14-10.65)^2}{10.65} + \frac{(54-49.07)^2}{49.07} + \frac{(26-28.52)}{28.52} + \frac{(12-14.41)^2}{14.41} + \frac{(35-35.2)^2}{35.2} + \frac{(22-20.46)^2}{20.46} + \frac{(9-10.34)^2}{10.34} + \frac{33-39.47}{39.47} + \frac{29-22.94}{29} + \frac{12-11.59}{11.59}$$

$= 6.446 < \chi^2_{6,0.05} = 12.59$

∴ not enough evidence to reject H_0

p-value: $P(\chi^2 > 6.446) \geq P(\chi^2 > 5.348) = 0.5$, so non reject

18) 200个人，不同 entrance，是否无差异

E	1	2	3
Number entering	83	61	56

H_0: each $p_i = p_j$, $i \neq j$

H_a: $p_i \neq p_j$, $i \neq j$

$E_i = \frac{200}{3} = 66.67$

$$\chi^2 = \sum_i \frac{O_i - E_i}{E_i} = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67}$$

$$= 6.19$$

$\chi^2_{2,0.05} = 5.991 < \chi^2 = 6.19$, so there is enough evidence to support H_a.

95% CI of E_1, $CI = \cancel{83 \pm \sqrt{np(1-p)}}$

$= \frac{83}{200} \pm Z_{0.05}\sqrt{\frac{p(1-p)}{n}}$

$= \frac{83}{200} \pm 1.96\sqrt{\frac{\frac{83}{200}\left(\frac{117}{200}\right)}{200}}$

$\approx \frac{83}{200} \pm 1.96(0.0348)$

$= (0.347, 0.483)$

注意：$Var\left(\frac{X}{n}\right) = \frac{1}{n^2} Var(X) = \frac{n(1-p)p}{n^2} = \frac{(1-p)p}{n}$

22)

H_0: $p_i = p_j$, $i \neq j$

H_a: $p_i \neq p_j$, $i \neq j$

$E_i = \frac{200}{7}$
$= 28.57$

$$\chi^2 = \sum_i \frac{(O_i - \hat{E}_i)^2}{\hat{E}_i} = \frac{(24-28.57)^2}{28.57} + \frac{(36-28.57)^2}{28.57} + \frac{(27-28.57)^2}{28.57} + \frac{(21-28.57)^2}{28.57}$$
$$+ \frac{(32-28.57)^2}{28.57} + \frac{(26-28.57)^2}{28.57} + \frac{(29-28.57)^2}{28.57} \approx 7.67$$

$\chi^2_{6,0.05} = 12.596 > \chi^2$, so there is not sufficient evidence to reject H_0.